

**A MAP-ONLY FRAMEWORK FOR PREDICTING RESIDUE PLACEMENT
CONFIDENCE DURING CRYO-EM MODEL BUILDING**

by
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TITLE: A Map-Only Framework for Predicting Residue Placement
Confidence During Cryo-EM Model Building

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DEDICATION

Dedicated to my younger self, who decided to go through this challenging life experience, and to everyone that helped me learn to see the world as fundamental particles and their interactions. Your guidance shaped this work, and the physician-scientist I hope to become.

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ABSTRACT

Structural biology seeks to determine the three-dimensional arrangement of atoms within biological macromolecules to understand their functions. Recent advances in cryogenic electron microscopy (cryo-EM) have enabled the routine determination of macromolecular structures at near-atomic resolution without the need for crystallization, making cryo-EM a central technique in modern structural biology. Despite these advances, the reconstructions produced by cryo-EM are not yet atomic models. Rather, they are three-dimensional density maps from which an atomic model must be subsequently constructed through model building.

Depending on the resolution of the map, the process of model building can be interpretive. A growing number of reconstructions now reach true atomic resolution, where the density is unambiguous and requires little interpretation. However, many reconstructions still fall within the 2.5–4 Å average-resolution range, where the challenges described below are present throughout. Although some regions of a density map contain well-defined features that permit unambiguous atomic placement, other regions exhibit weaker or less informative densities arising from molecular flexibility, conformational heterogeneity, limited local signal, or reconstruction uncertainty. Consequently, not all regions of a cryo-EM map can be interpreted with equal confidence, and identifying these differences is an essential component of accurate structure determination.

During cryo-EM model building, researchers must decide where to place atomic structures within the density map, often before any model exists (from crystallography or NMR). Local resolution is a measure that builders routinely consult when deciding how much of a map to interpret; however, it primarily reflects reconstruction reproducibility rather than how well the density constrains atomic positions. These are distinct questions, and the extent to which they diverge depends on the local-resolution estimator used. I introduce the reliability score, a map-only metric

computed from the local gradient magnitude of the normalized half-map average, which requires no fitted model. From the local gradient magnitude, a percentile-ranked reliability score was derived and binned into omit/caution/build terciles to provide actionable placement guidance. Validated against the Q-score, a cryo-EM-native, model-to-map measure of placement confidence, the reliability score tracks per-residue placement confidence across a cohort of 55 maps and identifies hard-to-place residues with a median per-map AUC of 0.82. It tracks EMRinger slightly more closely than it tracks the Q-score (cohort-median $\rho = +0.62$ versus $+0.53$). Against the coordinate corrections published by MEDIC, an independent residue-level detector of model errors, residues assigned to the omit zone are strongly enriched for subsequent correction. On the same structures, four local-resolution estimators predict placement confidence less consistently and differ markedly from one another in how well they do so. The advantage over the strongest estimator is small, and it does not survive reduction of both predictors to coarse labels. The utility of this tool weakens above roughly 4 Å, where the half-map average no longer varies finely enough to resolve placement difficulty at the residue scale, and is not supported above roughly 6 Å global resolution; therefore, the 2.5–4 Å band remains the primary operating regime for this metric.

The framework is implemented as cryoRIDGE (Reliability Inferred from Density Gradient Energy), a Python package that computes the reliability score and its omit/caution/build zones from a pair of unsharpened half-maps. This work argues that the map-only, gradient-magnitude-based metric termed reliability provides a more consistent basis for ranking placement difficulty during model building than local resolution, whose usefulness for this purpose varies substantially with the estimator chosen.

Keywords: cryo-EM, structural biology, model building, map quality, local resolution.

Full text available upon request.

VITA

Sarthak Mohanty earned his Bachelor of Science in Biochemistry from The University of Texas at San Antonio in August 2026. He began research in an organic chemistry group on campus, after he learned small-molecule X-ray crystallography with Dr. Hadi Arman and served as a teaching assistant for Dr. Arman.

He continued his research career in Dr. Maria Falzone’s laboratory, working with Dr. Falzone and Dr. Rahul Mojindra on the structure and function of phospholipase C epsilon ($\text{PLC}\epsilon$), a signaling protein frequently dysregulated in cancers. That work used cryogenic electron microscopy and fluorescence spectroscopy to study how phospholipases regulate cellular signaling pathways and how those mechanisms might eventually be targeted therapeutically. He was co-advised by Dr. Falzone and Dr. Arman. After graduation, he is pursuing an MD–PhD physician-scientist program.